P-01: PAH docking ranks and pocket contact

What do the stored PAH calculations show?

Materials and methods

Four original docking JSONs; six original distance XVGs; two original replicate-2 MM-GBSA DATs.

Source distance script and MD/rescoring condition notes; inputs are frozen local copies.

Recalculate ranks within tool and target; do not pool scoring scales.

Summarize strict distance <0.5 nm over each finite trace and extract reported rescoring means/SD.

Scope and working assumptions

The analysis compares stored docking scores, finite initialized MD traces and one selected replicate-2 MM-GBSA result. Contact is distance below 0.5 nm. Run initialization differs, frames are correlated and CB-Dock3 shares Vina scoring. Rescoring omits entropy and uses differing salt conventions. These quantities are computational comparisons, rather than experimental affinities, equilibrium residence times or selectivity measurements.

Results

35 docking scores; PahP first ranks: CB-Dock3: benzo_a_pyrene; Glide: anthracene; Vina: benzo_a_pyrene.

Contact fractions: pyrene: 19.80%, 100.00%, 100.00%; phenanthrene: 8.00%, 99.68%, 21.13%.

Replicate-2 MM-GBSA displayed-mean difference: 9.72 kcal/mol favouring pyrene.

Ligand	Run	Contact (%)	Last observed contact (ns)	Contact at end	Input data	Finished without more wet-lab data?
pyrene	1	19.8	23.34	False	Original distance XVG	Yes: stored computations
pyrene	2	100	100	True	Original distance XVG	Yes: stored computations
pyrene	3	100	100	True	Original distance XVG	Yes: stored computations
phenanthrene	1	7.999	8	False	Original distance XVG	Yes: stored computations
phenanthrene	2	99.68	100	True	Original distance XVG	Yes: stored computations
phenanthrene	3	21.13	21.12	False	Original distance XVG	Yes: stored computations

Interpretation

The docking methods give different first-ranked PahP ligands (CB-Dock3: benzo_a_pyrene; Glide: anthracene; Vina: benzo_a_pyrene). Pocket-contact fractions in runs 1-3 are pyrene: 19.80%, 100.00%, 100.00%; phenanthrene: 8.00%, 99.68%, 21.13%. Pyrene has the higher contact fraction in each run, and the selected replicate-2 MM-GBSA means favour pyrene by 9.72 kcal/mol.

References and reproducibility

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/md

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/3539358236841e3f95f3e9efdc6d954fe1a1eca2/tools/autodock-vina/results

https://github.com/Raminyazdani/igem-saarland-2026-docking-comparison/tree/04fd02b7d71f14314549e2e0887300ea54fd5035/tools

Complete inline code: tasks/P-01/P-01.ipynb. Numerical data and source/software provenance: data/output/P-01/.

Figures

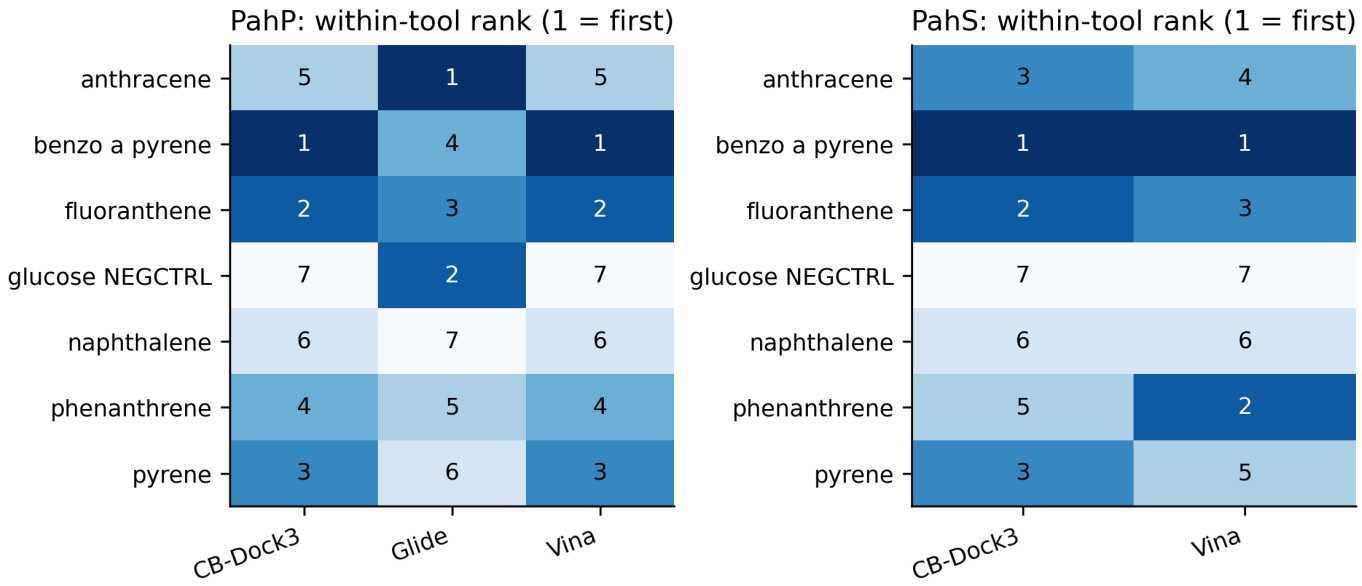


Figure 1. Within-tool/target ranks; 1 is first. Missing methods remain absent; scores are not pooled.

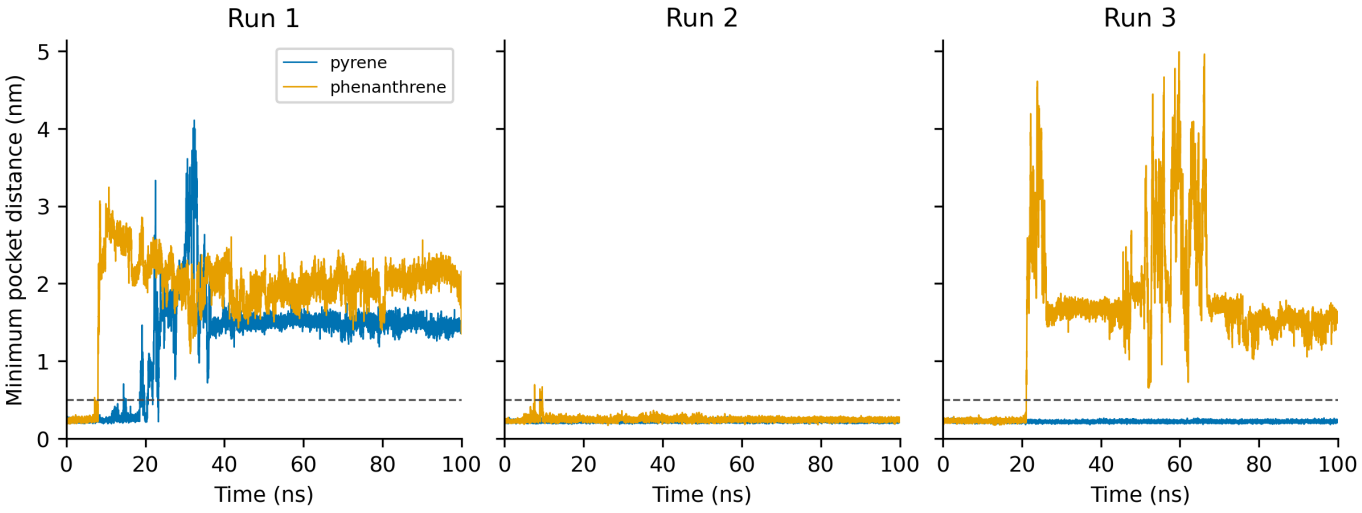


Figure 2. All six stored finite traces; dashed line is the strict 0.5 nm criterion. Run 1 initialization differs from runs 2-3.